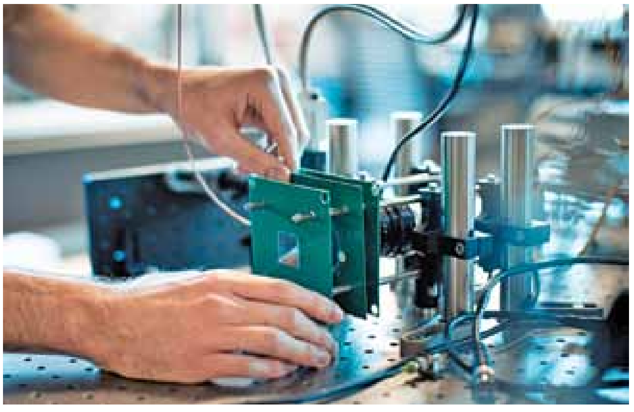
FAMU-FSU College of Engineering researchers are working to improve 3D printing technology by teaching machines to learn from one another—a great example of group intelligence. The researchers used a cloud platform to connect different printers, and then had the machines communicate data on accurate processing, reducing the time it took to prepare and calibrate them. They also constructed a mathematical model to better understand the printing process. The technology demonstrates that data generated from multiple production machines can be shared with each in a timely manner, and manufacturing can be done as an online service for meeting diverse market demands.

Smartphone cameras may soon capture 3D images

Stanford University researchers have developed a novel method for ordinary image sensors to sense light in three dimensions using lidar. The research team's lidar system



The lab-based prototype lidar system that the research team built, which successfully captured megapixel-resolution depth maps using a commercially available digital camera (Credit: https://news.stanford.edu)

was able to capture megapixel-resolution depth maps with a commercially accessible digital camera. Simply put, their all-new approach is simple and integrates into everyday cameras like cellphones. With this technology, cameras will be able to estimate the distance between objects, resulting in a three-dimensional image that can be viewed on smartphones. This way, smartphone cameras may be able to capture three-dimensional images (3D).

New gate-oxide tech can produce better transistors

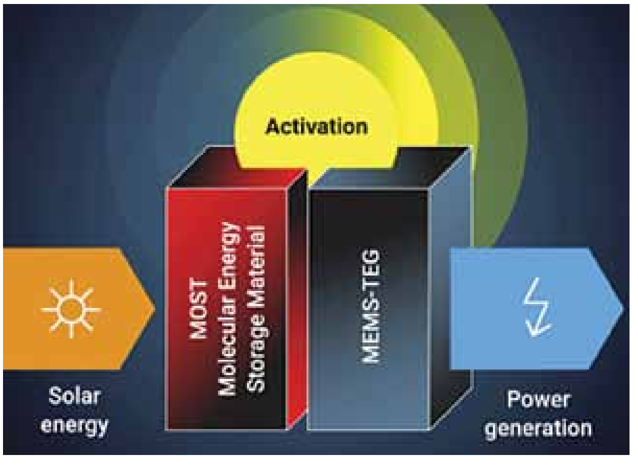
Engineers from the University of California, Berkeley have demonstrated a major breakthrough in the design of gate-oxides of transistors that could significantly reduce their energy consumption without sacrificing speed, size, or performance. They did this by leveraging negative capacitance to make better transistors with the help of engineered crystal structures. The research shows how negative capacitance may be obtained in an engineered crystal made up of a layered stack of hafnium oxide and zirconium oxide that is compatible with modern silicon transistors. They were able to show that their gate-oxide technology is better than commercially available transistors.

Speech tech for information in seven northeastern languages

IIT Guwahati researchers are developing a speech technology tool that will enable retrieval of healthcare related information in seven northeastern languages. This new technology uses spoken keyword spotting (KWS) in northeastern languages. The project's KWS systems will be able to detect a list of predefined terms in a given voice signal in one of the project's target languages. Modeling speech with state-of-the-art deep neural network techniques will be part of the endeavour. People in far-flung parts of Northeast India should be able to obtain healthcare-related information in their own languages as a result of this project.

Electricity on demand from solar energy stored up to 18 years

Researchers at Chalmers University of Technology in Sweden have devised an energy system that can capture solar energy, store it for up to eighteen years, and then release it as needed. By connecting the system to a thermoelectric generator, scientists have now succeeded in making



The concept of MOST (Credit: https://www.sciencedaily.com)